



420-1C1-DW PREVENTATIVE MONITORING · LEA.D8 · FALL 2026 · CLASS 11

Monitoring Tools 2: Installation, Updates and the Baseline

Lesson 3.3 Installing monitoring tools and applying updates

Lesson 3.4 Creating the reference database (baseline of normal activity)

Lab 2 summary

Friday, October 2, 2026 · 18:00-22:00 EDT (Eastern Daylight Time), online · Competency EZ01, elements 3.3 and 3.4

Instructor: Asen Grozdanov · Cyber.SoHo Educational Hub · Dawson College

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Today's plan

1

Overview

Where today sits in Topic 3; what Classes 6 to 8 supply

2.5 min

2

Lesson 3.3

Installing monitoring tools and applying updates: seven stages, ports, checks, hardening, update cycle, rollback

29 min

3

Lesson 3.4

Creating the reference database: what a baseline contains; building, cleaning, documenting it; thresholds

25 min

4

Lab 2 summary

What Saturday's Lab 2 asks and which of today's deliverables it reuses

2.5 min

5

Wrap-up and links

Closing points, dates, every external reference with its VirusTotal check

1 min

Where we are

Yesterday: mid-term (Topics 1-2, Lessons 3.1-3.2).

Today: 3.3 proper installation of updates; 3.4 efficient creation of the reference database.

Saturday, Oct 3: 3.5 configuring alerts, then Lab 2 starts.

Oct 10: Quiz 3. Oct 11: Lab 1 due. Oct 16: Lab 3.



The Val-des-Pins pilot: who is who

MON-01

lab monitoring server: VM (virtual machine), 4 vCPU, 8 GB RAM, 200 GB disk, Ubuntu Server 24.04 LTS

FW-01

perimeter firewall - syslog source, Class 7 priority 1

PC-ACCT-01

accounting workstation, Tier 1 - receives an agent

SW-01 / SW-02

core and workshop switches - syslog

SRV-FILE-01

file server, Tier 1 - receives an agent; the baseline of Lesson 3.4

NAS-01

backup storage - syslog, included by documented override

CNC-PC-01

machine computer - no software allowed by the vendor contract; documented gap

MAIL-CLOUD

hosted e-mail service (cloud)

Class 8 decision

Wazuh selected for the pilot (89.0 %), Graylog Open runner-up (69.0 %)

All names, systems and numbers are invented for the course (Legal Notice, sections 7.2 and 18.2). Software versions are those published in September 2026: check the official documentation before Lab 2.



3.3

Installing monitoring tools and applying updates

Competency element 3.3 - proper installation of updates.

Seven stages, the ports of the pilot, checks and hardening, three kinds of updates, the update cycle and rollback.



3.3.1 Installation is a procedure, not a download

- From a decision on paper (Class 8: Wazuh for the pilot) to a running system - the step where first deployments fail
- **A monitoring tool is different:** it receives a copy of what happens on every watched system (logons, file changes, firewall decisions)
- Installed carelessly, three things go wrong at once:
 - a tampered copy that sends your data elsewhere
 - a configuration that misses the events chosen in Lab 1
 - a tool that eats the disk, memory and processor of the lab VM (Virtual Machine) - the risk section 6.4 of the Legal Notice covers with a snapshot
- **Procedure:** a fixed sequence of stages, each with a check, that gives the same result every time
- **The word to remember: repeatable.** A colleague can rebuild the server from your written record



The seven stages of a proper installation

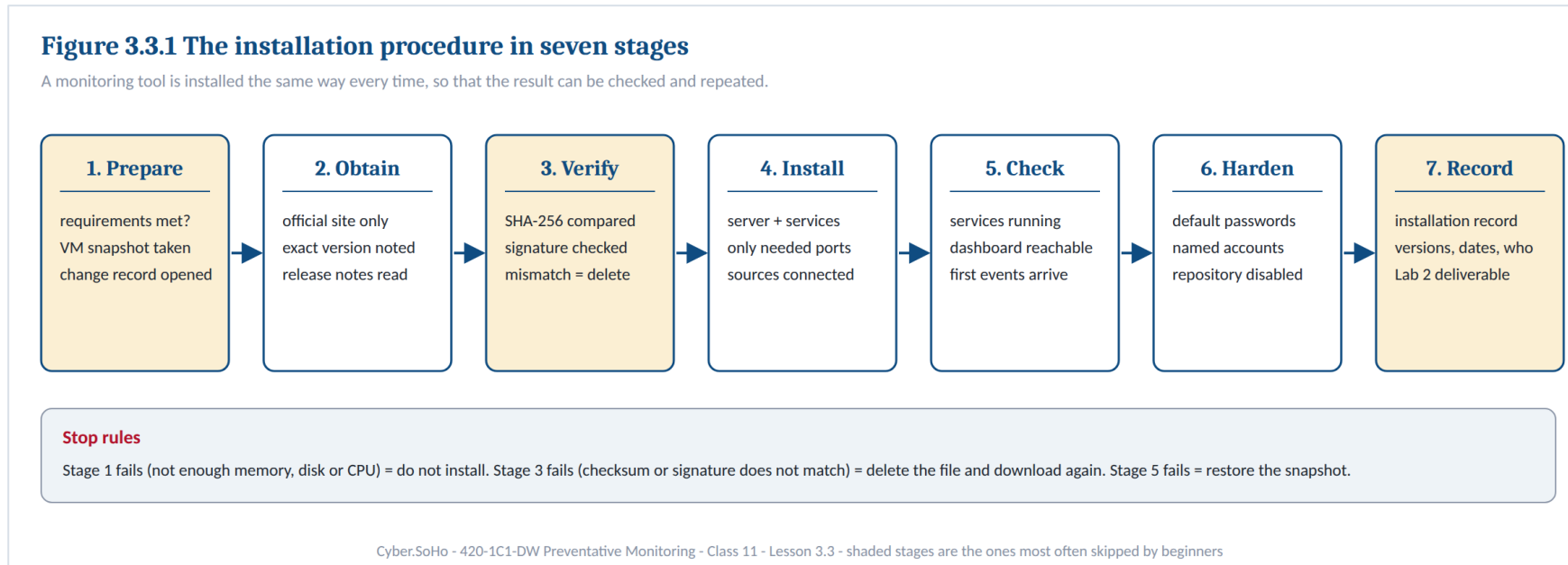


Figure 3.3.1 - Prepare, obtain, verify, install, check, harden, record. Shaded stages are the ones most often skipped.

- **Three stages are skipped most often:** preparing (nobody checks the memory), verifying (nobody compares the checksum), recording (nobody writes it down)
- **Each skipped stage removes a safeguard;** skipping all three gives an installation nobody can trust and nobody can rebuild
- These seven stages are the frame of Lab 2 (Section 4 of the documentation)

3.3.2 Before installing (1/2): requirements and official source

- **System requirements:** the minimum processor, memory, disk and operating system the publisher tested the tool with
- **Vendor table (all-in-one, 1 to 25 endpoints):** 4 vCPU, 8 GB RAM, 50 GB disk for 90 days of searchable alerts
- **MON-01 (Class 8):** 4 vCPU, 8 GB, 200 GB - pass, with room for the 90-day retention of need N-01
- **Write it down:** "required 4 / 8 / 50, available 4 / 8 / 200, pass" - the first row of the installation record
- **Operating system on the supported list:** Ubuntu Server 24.04 LTS (Long-Term Support) - five years of security updates
- **Official source only:** the publisher's own website or package repository - never a mirror from an advertisement, a forum or a video description
- **Note the exact version (4.14.7):** "the latest" is not a version - next week it will be something else



3.3.2 Before installing (2/2): checksum, snapshot, record

- **Checksum (SHA-256, Secure Hash Algorithm 256-bit):** a 64-character number computed from the whole file; one changed byte changes it completely
- **Compare the value you compute with the value the publisher published:** match = the file is the file
- **Mismatch = corrupted or tampered - same action either way:** delete, download again from the official source, never install
- **Digital signature:** the publisher signs the file with a private key; you verify with the public key (GPG). Proves origin and integrity at once - repositories do it automatically
- **Snapshot of the VM, named with the reason:** "MON-01 pre-install 4.14.7" - any failure undone in one action (Legal Notice, section 6.4)
- **Change record:** what will be installed, on which host, when, by whom, and the way back - a ticket in a company, a row in your record in the lab



Table 3.3.1 - Pre-installation checklist for MON-01

No	Check	How it is done in the lab	Evidence written in the record
1	System requirements met	Compare MON-01 (4 vCPU, 8 GB RAM, 200 GB disk, Ubuntu 24.04 LTS) with the vendor table for 1-25 endpoints (4 vCPU, 8 GB, 50 GB)	required 4 / 8 / 50, available 4 / 8 / 200, pass; OS on the supported list
2	Official source identified	Documentation page of the publisher; note the exact version (4.14.7)	URL of the page, version, date of download
3	Integrity verified	Compute the SHA-256 of the downloaded file and compare with the published value; check the repository signing key	computed and published values, 'match'
4	Snapshot taken	VirtualBox: Take snapshot 'MON-01 pre-install 4.14.7'	snapshot name, time
5	Change record opened	One row: what, on which host, when, who, way back (snapshot)	row number in the change log
6	Network isolation confirmed	MON-01 on the host-only lab network; no bridge to the home or campus network	adapter mode in VirtualBox
7	Time synchronised	MON-01 and the monitored VMs use the same NTP (Network Time Protocol) source; clocks within seconds	output of the time check

Full table with the Legal Notice references: [Table_3.3.1_Pre-installation_Checklist.csv](#) (companion files).

Terms to learn - Lesson 3.3

1

System requirements

The minimum processor, memory, disk and operating system versions the publisher tested the tool with.

2

Official source

The publisher's own website or package repository; the only place a tool is downloaded from.

3

Checksum (SHA-256)

A 64-character hexadecimal number computed from the whole file; any change to the file changes it.

4

Digital signature

A checksum encrypted with the publisher's private key; verifying it proves origin and integrity at once.

5

Snapshot

A saved state of a virtual machine that can be restored in one action; taken before every installation and update.

6

Change record

The written entry that says what is changed, on which host, when, by whom and how it can be undone.



3.3.3 Installing the server: three components and one agent

- 1 Manager (server)**
receives events, decodes them, matches them against rules, produces alerts
- 2 Indexer**
stores the alerts and makes them searchable - 90 days of history queried in seconds
- 3 Dashboard**
the web interface: search, graphs, agent management (port 443)
- 4 Agent**
small program on a monitored host: reads logs, checks file integrity, inventories the system, sends to the manager over an encrypted channel

All three central components run on MON-01 (all-in-one); in a large organization they are spread over several servers.

The installation assistant (one script, administrator rights)

1. adds the official repository and imports its signing key
2. installs manager, indexer and dashboard
3. generates the certificates they use between them
4. starts the services
5. prints the initial administrator password

Then: disable the package repository, so that an ordinary operating system update cannot upgrade the platform by accident (see 3.3.5).

Table 3.3.2 - Ports MON-01 accepts in the pilot

Port / protocol	Who connects	Purpose	Note for the pilot
1515 / TCP	new agents (PC-ACCT-01, SRV-FILE-01)	Enrolment: the agent presents its name and receives its unique key	Can be closed between enrolments
1514 / TCP	enrolled agents	Events from the agent to the manager, encrypted with the agent key	Stays open; one connection per agent
514 / UDP	FW-01, SW-01, NAS-01	Syslog messages from devices that cannot run an agent (agentless)	No encryption: acceptable in the isolated lab network only
443 / TCP	the analyst's browser	Web dashboard over HTTPS	Self-signed certificate at first; replacement planned before Lab 3
55000 / TCP	administration tools on MON-01	Server API used by the dashboard and scripts	Restricted to MON-01 itself; not reachable from other hosts
9200 / TCP	the manager and the dashboard	Indexer (search and storage) API	Internal to the all-in-one host; never exposed

Enrolment on 1515/TCP (the agent gets its key), then events on 1514/TCP encrypted with that key; syslog on 514/UDP carries no encryption - acceptable only in the isolated lab network. Every open port is recorded with its reason.



Connecting the sources in the Class 7 order

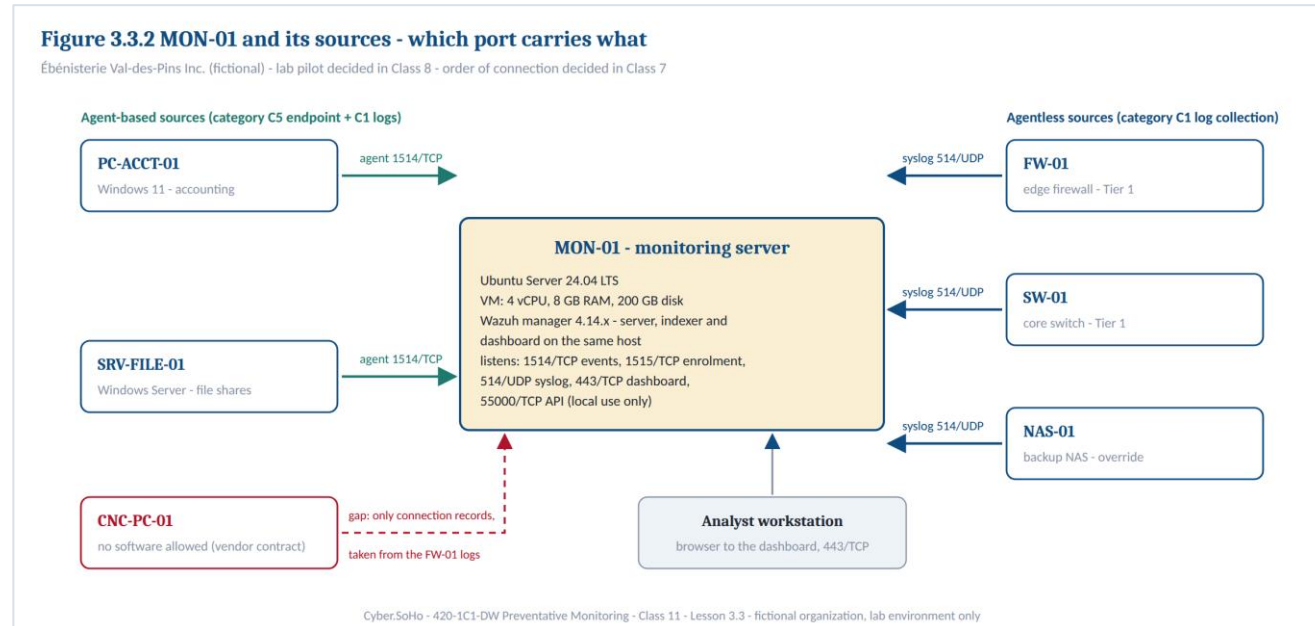


Figure 3.3.2 - MON-01 and its sources, with the port each connection uses.

- **Order = the Class 7 priority list:** FW-01, PC-ACCT-01, SW-01, SRV-FILE-01, then CNC-PC-01 and NAS-01
- **Two agents (PC-ACCT-01, SRV-FILE-01):** logs, file integrity, software inventory. Three agentless devices: syslog pointed at MON-01
- **CNC-PC-01 receives nothing (vendor contract):** written as "no agent, connection records taken from FW-01" - a documented gap is a decision; an undocumented gap is a blind spot

3.3.4 Checking and hardening a fresh installation

- Verify against observable evidence, never against "installation finished":
 - services running - manager, indexer, dashboard all "active"
 - dashboard answers on port 443 and accepts the administrator login
 - every enrolled agent shows "active" with the installed version
 - first events arrive: a test logon on PC-ACCT-01 visible within a minute, a syslog message from FW-01 found in the search
- Three passes and the fourth fails = a server that runs and monitors nothing - worse than no server (false confidence)
- **Hardening** makes the fresh installation harder to misuse:
 - replace generated passwords (administrator and internal accounts) - stored in a password manager
 - a named read-only account for daily checks (least privilege)
 - record the self-signed certificate; plan its replacement before Lab 3
 - disable the package repository - the version changes only when you decide



Table 3.3.3 - Post-installation checks and hardening

Stage	Action	Expected result (evidence)	If it fails
Check	Service status of manager, indexer, dashboard	all three 'active (running)'	Read the service log; restore the snapshot if the cause is not obvious
Check	Open the dashboard on https://MON-01	login page answers; administrator login works	Verify port 443 rule and the dashboard service
Check	Agent list	PC-ACCT-01 and SRV-FILE-01 'active', version 4.14.7	Check port 1514/1515, the manager address in the agent configuration, the key
Check	First events arrive	a test logon on PC-ACCT-01 visible within one minute; a syslog message from FW-01 found in the search	The server runs but monitors nothing: check the source configuration before continuing
Harden	Replace generated passwords (admin, API users)	new strong passwords stored in the password manager	Never leave a generated password in a text file on the desktop
Harden	Create a named read-only account for daily checks	account exists; cannot delete rules or agents	Least privilege: the daily account is not the administrator
Harden	Record the certificate status	'self-signed, accepted in the lab, replacement planned'	A warning that is not understood is a warning that will be ignored later
Harden	Disable the package repository	an operating system update no longer changes the platform version	Updates happen only when you apply them (3.3.5)
Record	Installation record completed	every stage has a row: what, when, who, result	An installation without a record cannot be rebuilt or audited

CIS (Center for Internet Security) Benchmarks give the longer hardening list for the operating system underneath.



3.3.5 Three kinds of updates

- **Software update:** changes the program itself - manager, indexer, dashboard or agent
 - versions carry three numbers, e.g. 4.14.8: MAJOR (large, possibly incompatible changes) . MINOR (new features) . PATCH (fixes only) - semantic versioning, semver.org
- **Content update:** changes the detection rules, decoders and lists - not the program; issued often, sometimes weekly, because attacks change faster than software
- **Operating system update** of the monitoring server itself - a different product with its own cycle
- **Confusing the three kinds is a common source of trouble:** each has its own cadence and its own way back



It holds copies of everyone's logs and has a network path to every monitored host

Why update at all? The monitoring server is a high-value target

CVE

Common Vulnerabilities and Exposures - a publicly known weakness gets an identifier

www.cve.org [VirusTotal check](#)

NVD

National Vulnerability Database (NIST) - the weakness is scored

nvd.nist.gov [VirusTotal check](#)

CISA KEV

Known Exploited Vulnerabilities catalogue - attackers are using it: an unpatched server is the easiest door in the building

www.cisa.gov/known-exploited-vulnerabilities-catalog [VirusTotal check](#)

CCCS ITSM.10.096

Canadian Centre for Cyber Security: patching is action item 2 of the top ten

cyber.gc.ca/en/guidance/top-10-it-security-action-items-no2-patch-operating-systems-and-applications-itsm10096 [VirusTotal check](#)

NIST SP 800-40 Rev. 4

Patch management is a planned, recurring activity - not an emergency

doi.org/10.6028/NIST.SP.800-40r4 [VirusTotal check](#)

Both guidance documents say the same thing: decide how quickly each kind of update is applied, then do it on schedule.



The update cycle

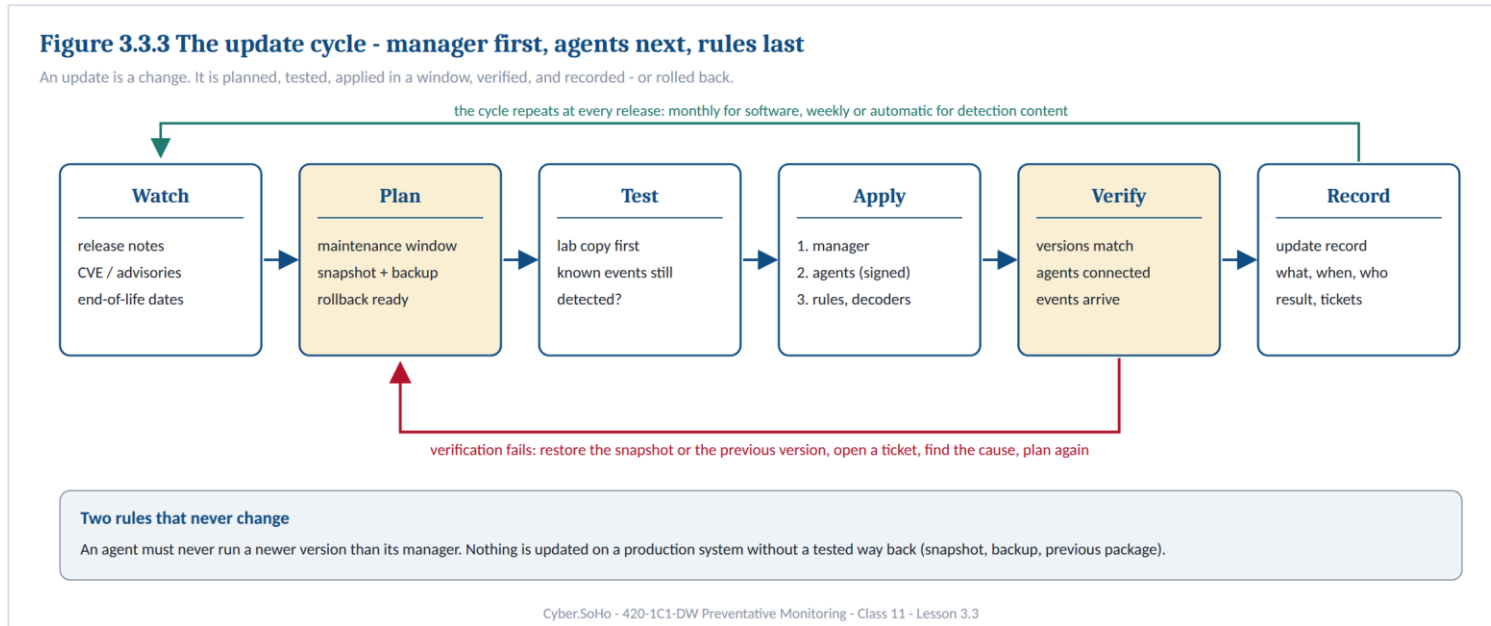


Figure 3.3.3 - The update cycle, with the rollback path when verification fails.

- **Watch the release notes and advisories:** patch (fixes only) or minor version (new features, possibly new requirements)?
- **Plan a maintenance window agreed with the owner (Val-des-Pins: Sunday morning) and take a snapshot before it opens;** in a small organization the snapshot is the test environment
- Apply in a fixed order, verify each component against a stated expectation, record the result

Order, verification and rollback

- **Fixed order:** manager first, agents next, detection content last
 - an agent must never be newer than its manager - the manager may not understand what a newer agent sends
 - agents are updated from the manager: a signed package is pushed to each one and it restarts
 - content is refreshed last, once the software is stable; the restart it needs is tested
- **Verify each component** as after the installation: version, status, and events still arriving
- **Rollback** = return the component to its state before the change: the snapshot for the whole server, the previous package for one agent
 - immediately, before the next host is touched; the failure gets a ticket so the cause is found later, calmly
- **Two rules:** no agent newer than its manager, and no change without a tested way back



Table 3.3.4 - Update types, order and cadence for the pilot

Kind of update	What changes	Example	Order in the window	Cadence for the pilot	Way back
Software - patch (x.y.Z)	fixes only, no new feature	4.14.7 -> 4.14.8	manager, then agents	next monthly window after the release notes are read	snapshot / previous package
Software - minor (x.Y.0)	new features, possibly new requirements	4.14 -> 4.15	manager, then agents	after a test on the lab copy; planned window	snapshot
Software - major (X.0.0)	large, possibly incompatible changes	4.x -> 5.0	planned as a project	only after the upgrade guide is read and tested	snapshot + full backup
Content - rules, decoders, lists	detection knowledge, no program change	new rule for a known attack	last, after the software is stable	weekly, or automatic where the tool allows	previous content package
Operating system of MON-01	security fixes of Ubuntu	kernel or library patch	separate window	security updates automatically within a week; reboot in the window	snapshot
Agent on monitored hosts	the agent program	4.14.7 -> 4.14.8	after the manager, one host at a time	same window as the manager	previous agent package

Operating system security updates automatically within a week; platform patch releases in the next monthly window; minor versions after a test in the lab; detection content weekly or automatically.



Legal Notice checkpoints for Lesson 3.3

Section 6.4 - laboratory environment

Install tools only inside your Authorized Lab Environment, isolated from any other network, with a snapshot taken first.

Section 6.6 - software licences

You are responsible for the licence of every tool you run; use only official, licensed software.

Section 14.2 and Annex C - links and downloads

Download only from the publisher's official site; verify the checksum; open every link through its VirusTotal check first.

Section 7.2 - monitoring data

Every log the tool collects is treated as personal information: kept in the lab, kept to the minimum, deleted after grading.

Full text: [cyber-soho.github.io/prevmon - Legal Notice, Terms of Use and Acceptable Use Policy \(v2.0\)](https://cyber-soho.github.io/prevmon - Legal Notice, Terms of Use and Acceptable Use Policy (v2.0)) [VirusTotal check](#)



Videos for Lesson 3.3



Mitigation Techniques - CompTIA Security+ SY0-701 - 2.5 (Professor Messer, 6:51)

Video: <https://www.youtube.com/watch?v=Fc8ZJfmapbl> VirusTotal check · Course page: professormesser.com/security-plus/sy0-701/sy0-701-video/mitigation-techniques-sy0-701/ VirusTotal check

Patching on a schedule, emergency patches, and why monitoring and least privilege go together with updates. Watch the first three minutes for patching, the middle part for monitoring and SIEM.



Vulnerability Remediation - CompTIA Security+ SY0-701 - 4.3 (Professor Messer, 8:45)

Video: <https://www.youtube.com/watch?v=P9xakfmX70c> VirusTotal check · Course page: professormesser.com/security-plus/sy0-701/sy0-701-video/vulnerability-remediation-sy0-701/ VirusTotal check

Scheduled and unscheduled patches, exceptions when a patch cannot be applied, and verifying after a patch that it really removed the weakness: exactly the verify stage of Figure 3.3.3.

Open the VirusTotal check first, then the link (Legal Notice, section 14.2 and Annex C). Third-party content used for identification and educational purposes only (section 5.2).



Real-life example 3.3 (1/2): installing the pilot on MON-01

- **Task:** Install the Wazuh manager on MON-01, connect the six Val-des-Pins sources in the Class 7 order, harden the server, and one week later apply the patch 4.14.7 to 4.14.8 in a maintenance window, with verification and rollback. Produce the Installation and Update Record.
- **Relation to the lecture:** the seven stages of Figure 3.3.1, the ports of Table 3.3.2, the checks of Table 3.3.3 and the update order of Figure 3.3.3 - on the server chosen in Class 8, for the sources prioritized in Class 7
- **Deliverable:** the Installation and Update Record (Table 3.3.5, CSV) = Section 4 of the Lab 2 documentation; Lab 3 (October 16) reuses it
- **Saturday, October 3, 14:05 - Lab 2 starts:** MON-01 passes the requirements (4 / 8 / 200 against 4 / 8 / 50); snapshot "MON-01 pre-install 4.14.7"
- **Download from the official documentation page, compute the SHA-256:** the first comparison is made against a tampered copy on purpose (MISMATCH), the genuine file matches - only then the assistant runs



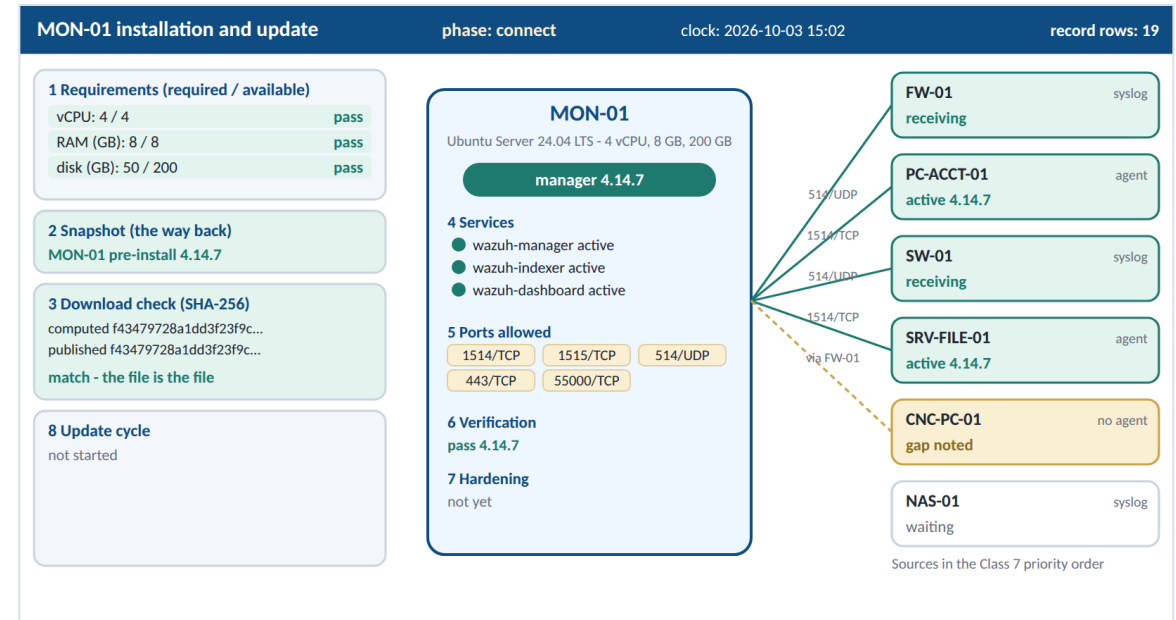
Real-life example 3.3 (2/2): from first events to the first patch

- **Twenty minutes later:** three services "active", dashboard on port 443 (self-signed warning written down), five firewall rules with their reasons, manager version verified
- **Sources in the Class 7 order:** FW-01, SW-01, NAS-01 over syslog (first messages seen); PC-ACCT-01 and SRV-FILE-01 enrol on 1515 and send on 1514 ("active 4.14.7"); CNC-PC-01 recorded as a gap - 23 rows in the record
- **Hardening:** new passwords, named read-only account, certificate status recorded, repository disabled
- **Sunday, October 11, 09:00 - maintenance window:** release notes 4.14.8 (patch, fixes only), snapshot "MON-01 pre-update 4.14.8", central components updated and verified
- **Agent push to PC-ACCT-01 fails verification** (still 4.14.7, disconnected > 5 min): rolled back to 4.14.7 at once, ticket 2026-0411 opened
- **SRV-FILE-01 verifies at 4.14.8; rules and decoders updated last;** the record closes at 33 rows - what was done, when, by whom, with which result



Code example 3.3 (1/2): the installation and update simulator

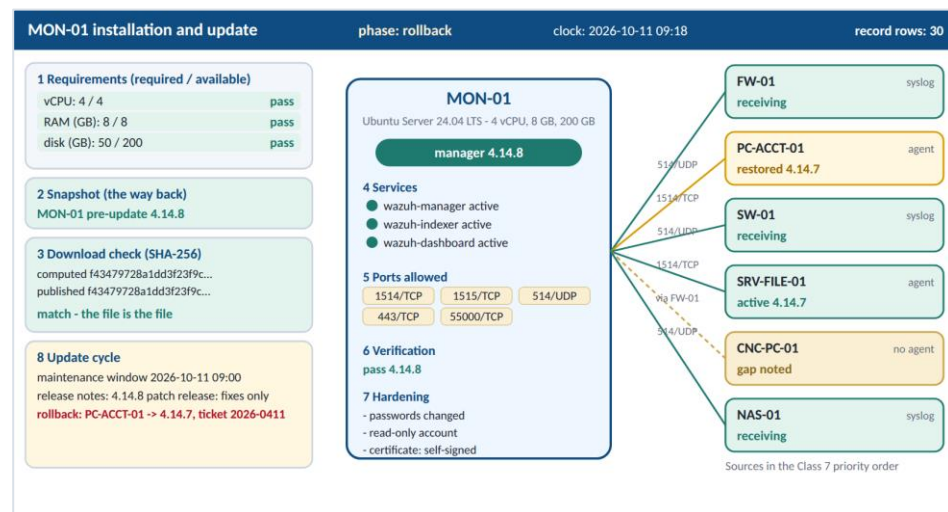
- **Program:** class11_install_update.py (Python 3, standard library; a comment line before every code line) - mirrored step by step by the simulator in the lecture page
- **Task:** simulate the seven stages and the update cycle, print the same trace, write the Installation and Update Record as CSV. Nothing real is installed
- **One function per stage:** check_requirements, snapshot, verify_download (SHA-256), install_manager, open_ports, connect_sources, harden, verify, rollback, apply_update
- **The picture shows the simulator after step 19:** requirements passed, snapshot taken, checksum matched, manager 4.14.7 with its ports, sources connected in the Class 7 order, CNC-PC-01 as a gap
- **Buttons in the page:** Start / Pause (resumes where it stopped), Restart, Step, Copy code, downloads (trace, visualization, program, simulator)



Code example 3.3 (2/2): the update with rollback, in the trace

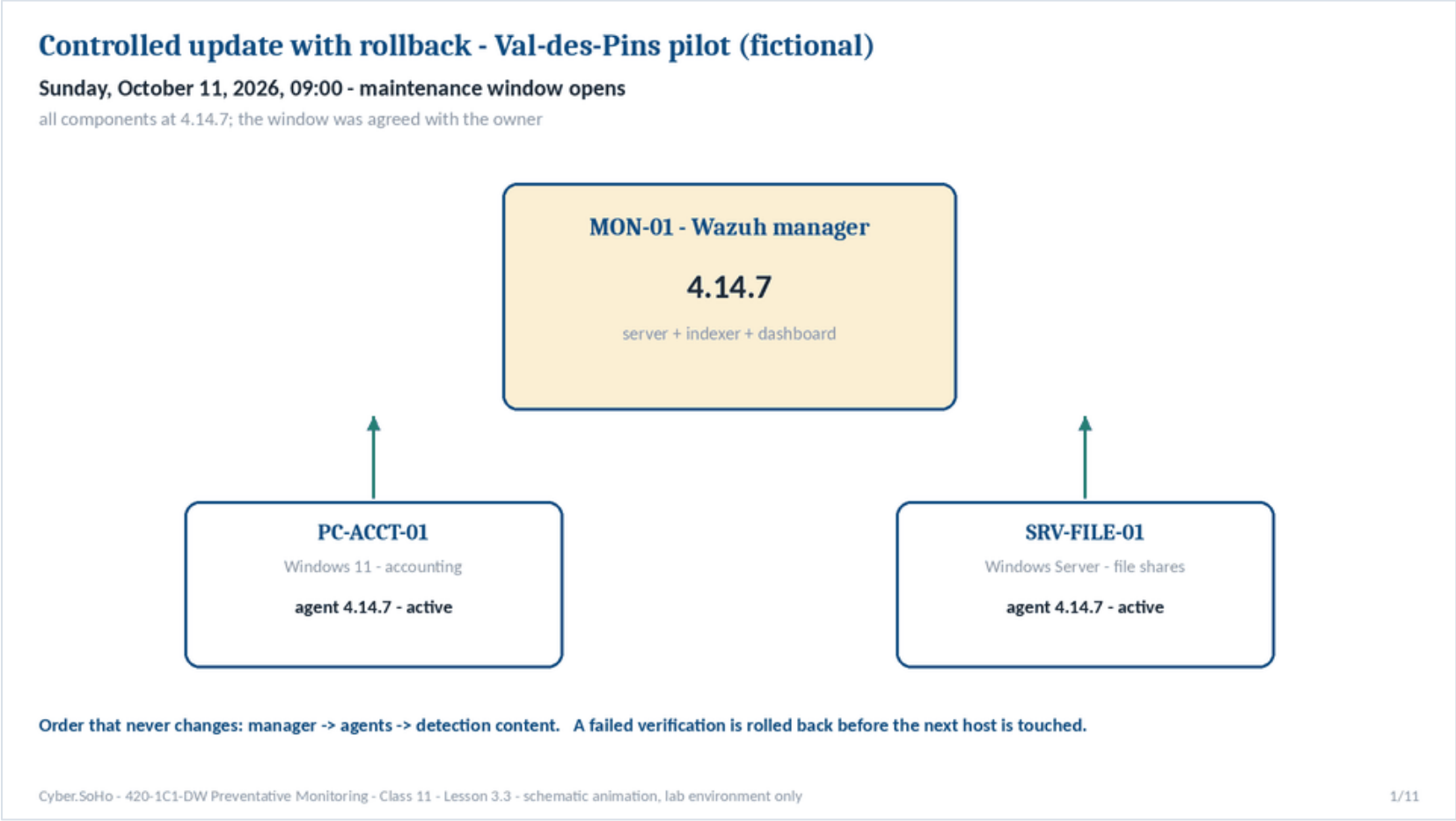
```
[24] 09:00 update   release notes read notes 4.14.7 -> 4.14.8
[25] 09:03 prepare MON-01      take VM snapshot
[26] 09:06 update   MON-01      update central components
[27] 09:09 verify   MON-01      check version and status
[28] 09:12 update   PC-ACCT-01  push agent update
[29] 09:15 verify   PC-ACCT-01  check version and status
[30] 09:18 rollback PC-ACCT-01  restore previous version
[31] 09:21 update   SRV-FILE-01  push agent update
[32] 09:24 verify   SRV-FILE-01  check version and status
[33] 09:27 update   MON-01      update ruleset and decoders
```

```
-> reviewed 4.14.8 patch re
-> done MON-01 pre-update 4
-> updated 4.14.8 maintenanc
-> pass 4.14.8 expected 4.14
-> sent 4.14.8 signed packag
-> FAIL 4.14.7 expected 4.14
-> restored 4.14.7 ticket op
-> sent 4.14.8 signed packag
-> pass 4.14.8 expected 4.14
-> updated content update,
```



- Manager first (line 26), verified before any agent (27)
- PC-ACCT-01 fails verification (29) - rolled back at once (30), ticket opened
- SRV-FILE-01 updated and verified (31-32); rules and decoders last (33)
- Every line is one row of the CSV record: 33 rows, the Lab 2 deliverable

Animation 3.3 - Controlled update with rollback



Animation 3.3 - The controlled update of Sunday, October 11, from the release notes to the record, including the failed verification of PC-ACCT-01 and its rollback.



Table 3.3.5 - Installation and Update Record (1/3)

No	Timestamp	Phase	Component	Action	Result	Version
1	2026-10-03 14:08	prepare	MON-01	check vcpu	pass	
2	2026-10-03 14:11	prepare	MON-01	check ram_gb	pass	
3	2026-10-03 14:14	prepare	MON-01	check disk_gb	pass	
4	2026-10-03 14:17	prepare	MON-01	take VM snapshot	done	
5	2026-10-03 14:20	obtain	installer	compare SHA-256	MISMATCH	
6	2026-10-03 14:23	obtain	installer	compare SHA-256	match	
7	2026-10-03 14:26	install	MON-01	run installation assistant	installed	4.14.7
8	2026-10-03 14:29	install	MON-01	start services	running	4.14.7
9	2026-10-03 14:32	install	MON-01	allow 1514/TCP	rule added	
10	2026-10-03 14:35	install	MON-01	allow 1515/TCP	rule added	
11	2026-10-03 14:38	install	MON-01	allow 514/UDP	rule added	

Produced by class11_install_update.py; the Note column is in the CSV file (companion files).



Table 3.3.5 - Installation and Update Record (2/3)

No	Timestamp	Phase	Component	Action	Result	Version
12	2026-10-03 14:41	install	MON-01	allow 443/TCP	rule added	
13	2026-10-03 14:44	install	MON-01	allow 55000/TCP	rule added	
14	2026-10-03 14:47	verify	MON-01	check version and status	pass	4.14.7
15	2026-10-03 14:50	connect	FW-01	point syslog to MON-01	receiving	
16	2026-10-03 14:53	connect	PC-ACCT-01	install and enrol agent	active	4.14.7
17	2026-10-03 14:56	connect	SW-01	point syslog to MON-01	receiving	
18	2026-10-03 14:59	connect	SRV-FILE-01	install and enrol agent	active	4.14.7
19	2026-10-03 15:02	connect	CNC-PC-01	no agent (vendor contract)	gap noted	
20	2026-10-03 15:05	connect	NAS-01	point syslog to MON-01	receiving	
21	2026-10-03 15:08	harden	MON-01	change default passwords	done	
22	2026-10-03 15:11	harden	MON-01	create named read-only account	done	

Rows 15 to 20 are the six sources; row 19 is the documented gap.



Table 3.3.5 - Installation and Update Record (3/3)

No	Timestamp	Phase	Component	Action	Result	Version
23	2026-10-03 15:14	harden	MON-01	record certificate status	self-signed	
24	2026-10-11 09:00	update	release notes	read notes 4.14.7 -> 4.14.8	reviewed	4.14.8
25	2026-10-11 09:03	prepare	MON-01	take VM snapshot	done	
26	2026-10-11 09:06	update	MON-01	update central components	updated	4.14.8
27	2026-10-11 09:09	verify	MON-01	check version and status	pass	4.14.8
28	2026-10-11 09:12	update	PC-ACCT-01	push agent update	sent	4.14.8
29	2026-10-11 09:15	verify	PC-ACCT-01	check version and status	FAIL	4.14.7
30	2026-10-11 09:18	rollback	PC-ACCT-01	restore previous version	restored	4.14.7
31	2026-10-11 09:21	update	SRV-FILE-01	push agent update	sent	4.14.8
32	2026-10-11 09:24	verify	SRV-FILE-01	check version and status	pass	4.14.8
33	2026-10-11 09:27	update	MON-01	update ruleset and decoders	updated	

Section 4 of the Lab 2 documentation is a record of exactly this shape, for your own lab VM.



Flash cards - Lesson 3.3

What are the seven stages of a proper installation, in order?

Prepare, obtain, verify, install, check, harden, record. Preparing, verifying and recording are the three most often skipped.

What does a matching SHA-256 checksum prove, and what does a mismatch tell you to do?

It proves the downloaded file is byte-for-byte the file the publisher published. A mismatch means: delete the file and download it again from the official source; never install it.

In which order are the parts of a monitoring platform updated, and why?

Manager first, agents next, detection content last. An agent must never be newer than its manager, because the manager may not understand what a newer agent sends.

Name the three kinds of updates a monitoring tool receives.

Software updates (major, minor, patch versions of the program), content updates (rules, decoders, lists), and operating system updates of the server that hosts it.

What must exist before any update is applied, and what is done when verification fails?

A tested way back: a snapshot or the previous package. When verification fails, the component is rolled back immediately, a ticket is opened, and the cause is found later.



Self-check 3.3 (1/3)

MCQ - one answer

Q1. MON-01 has 4 vCPU, 8 GB of RAM and 200 GB of disk. The vendor's table for 1 to 25 endpoints asks for 4 vCPU, 8 GB and 50 GB. What is the correct first row of the installation record?

- A) "Required 4 / 8 / 50, available 4 / 8 / 200, pass" - proceed to the snapshot.
- B) "Disk oversized" - reduce the virtual disk to 50 GB before installing.
- C) "RAM equal to the minimum" - fail, add memory before installing.
- D) No row is needed: requirements are checked only after the installation.

Hint: Compare each value with 'at least' in mind, and remember that the record is written before anything is installed.

MCQ - one answer

Q2. You download the installation file from the official site and compute its SHA-256. The value differs from the one published on the site. What do you do?

- A) Install anyway; small differences are normal after a download.
- B) Delete the file and download it again from the official source; install only when the values match.
- C) Change the published value in your record to match the computed one.
- D) Run the installer with administrator rights so that the checksum is recomputed.

Hint: A checksum comparison has only two outcomes, and one of them has only one acceptable action.



Self-check 3.3 (2/3)

MCQ - one answer

Q3. During the update window the manager is at 4.14.8 and you are about to update the agents. Which statement is correct?

- A)** Agents may be updated before the manager as long as they are restarted.
- B)** Agents must never run a version newer than the manager, so the manager is updated first and the agents follow.
- C)** Only the dashboard needs to be updated; agents update themselves automatically at every restart.
- D)** Agents are updated by re-installing the operating system of each host.

Hint: Think about who must understand whom.

MCQ - one answer

Q4. Which piece of evidence shows that the installation monitors something, rather than merely running?

- A)** The three services report 'active'.
- B)** The dashboard accepts the administrator login on port 443.
- C)** A test logon on PC-ACCT-01 appears in the dashboard within a minute.
- D)** The installer printed 'Installation finished'.

Hint: Three of the checks concern the server itself; one concerns the data.



Self-check 3.3 (3/3)

MSQ - select 3

Q5. Which three actions belong to the preparation stage, before any installation starts? (select 3)

- A) Compare the server with the published system requirements.
- B) Take a named snapshot of the virtual machine.
- C) Change the default administrator password.
- D) Open a change record that says what, where, when, who and the way back.
- E) Update the detection rules and decoders.
- F) Create a named read-only account for daily checks.

Hint: Preparation is everything that happens before the installer runs.

MSQ - select 3

Q6. PC-ACCT-01 fails verification after its agent update (it reports 4.14.7 and shows 'disconnected'). Which three actions are part of the correct response? (select 3)

- A) Roll PC-ACCT-01 back to 4.14.7 by re-installing the previous package.
- B) Open a ticket so the cause is found after the window.
- C) Roll the manager back to 4.14.7 as well, to keep everything equal.
- D) Continue with SRV-FILE-01: one failure does not stop the verification of the others.
- E) Delete the snapshot, since the manager update succeeded.
- F) Update the detection content first, then retry the agent.

Hint: Rollback is immediate; investigation is later; the other hosts are not held hostage.



3.4

Creating the reference database (baseline of normal activity)

Competency element 3.4 - efficient creation of the reference database.

What a baseline contains, how it is built over two weekly cycles, cleaned after the owner review, documented, versioned, and turned into thresholds.



3.4.1 What is normal here? The reference database

- The tool now receives a copy of everything - every logon on SRV-FILE-01, every decision of FW-01, every change to a watched file
- **Before any alarm, one question:** what is normal here? 35 logons on the file server at 22:00 are alarming in a workshop that closes at five, unremarkable in a hospital
- **The tool cannot know which one it is in:** somebody measures it and writes it down - that written measurement is the reference database, called the baseline in most tools
- **Baseline:** a documented description of the normal activity of a system, a network or an account, measured over a representative period, with enough detail to compare new activity against it
- **Anomaly-based detection (Topic 5):** an anomaly is activity that departs from the baseline by more than an agreed margin - no baseline, no margin
- **NIST SP 800-94:** profiles of normal behaviour built over a training period, significant deviations flagged. **NIST SP 800-92:** log analysis is only as good as the understanding of what logs normally contain

3.4.1 Created once, maintained always

- **Competency element 3.4:** "efficient creation of the reference database"; element 5.1 (Class 17): its "comprehensive update" - the two belong together
- **Networks change:** a new employee, a new server, a new backup schedule all move the line of normal
- A baseline that is not reviewed drifts in both directions:
 - it raises alarms about activity that is now ordinary
 - it stops noticing activity that is now suspicious, because the margin was set for a busier time
- So the Lab 2 Baseline Record carries a version number, a date and a review date from day one
- **Review date for the pilot:** Friday, October 16, when Lesson 6.1 uses it



3.4.2 The four families of reference values

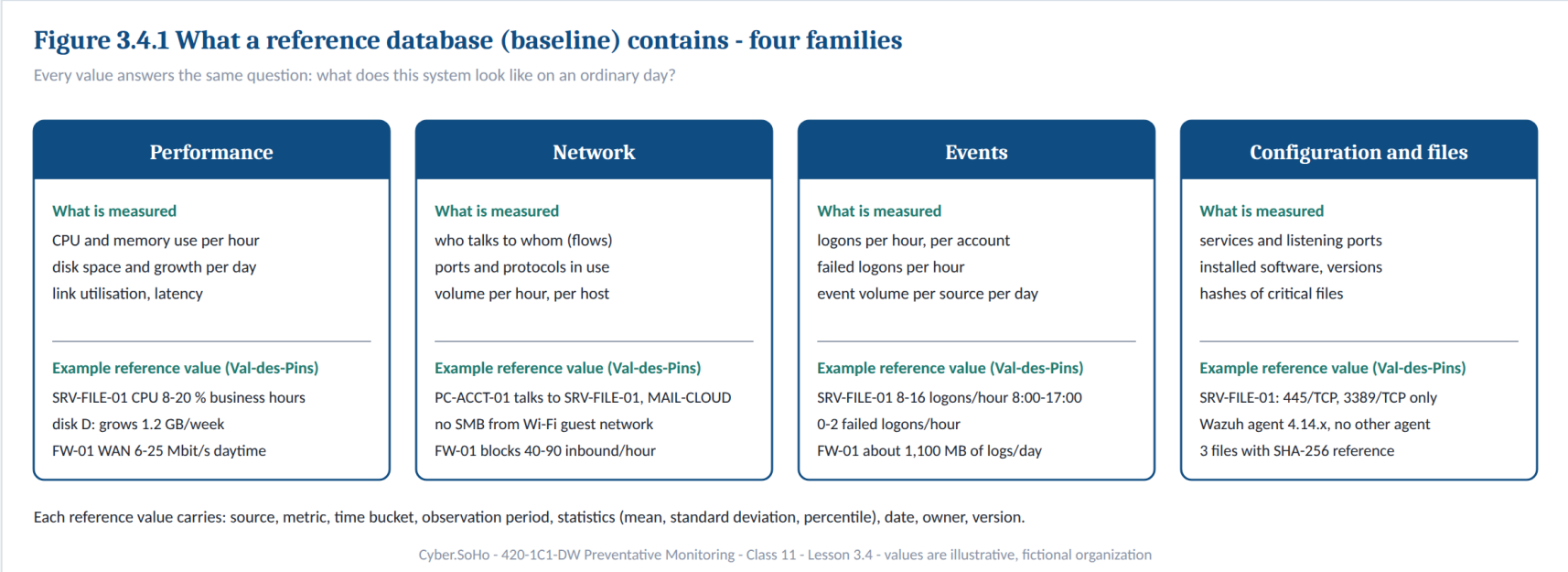


Figure 3.4.1 - Performance, network, events, configuration and files - with Val-des-Pins example values.

- **Performance:** how busy a system is (CPU, memory, disk growth, link utilisation) - a security problem often shows first as a performance change
- **Network:** who talks to whom, over which ports, how much per hour. **Events:** logons, failed logons, errors, event volume per source (protects the Class 7 envelope: FW-01 about 1,100 MB/day)
- **Configuration and files:** services, listening ports, installed software, SHA-256 of files that must not change silently (the FIM baseline; NIST SP 800-128: the baseline configuration)



Table 3.4.1 - The four families in the pilot

Family	Measured by (pilot)	Reference value for Val-des-Pins	Bucket	Used by
Performance	agent inventory (system data), SNMP counters on SW-01	SRV-FILE-01 CPU 8-20 % in business hours; disk D: grows about 1.2 GB per week	weekday business hours / off hours	Lesson 6.2 (impact of monitoring), Topic 5
Network	FW-01 connection records, SW-01 counters	PC-ACCT-01 exchanges traffic with SRV-FILE-01 and the cloud mail service only; no SMB from the guest Wi-Fi; FW-01 blocks 40-90 inbound attempts per hour	weekday hour / weekend	Lesson 3.5 (alerts), Topic 4
Events	agent log collection, syslog from FW-01	SRV-FILE-01 successful logons per hour (Table 3.4.3); 0-2 failed logons per hour; FW-01 about 1,100 MB of logs per day	weekday hour / weekend; per day for volume	Lesson 3.5 (alerts), Topic 5
Configuration and files	agent inventory, security configuration assessment, file integrity monitoring	SRV-FILE-01 listens on 445/TCP and 3389/TCP only; Wazuh agent 4.14.x is the only agent; 3 files with SHA-256 reference (Table 3.4.4)	not time-based: current state vs reference	Lesson 6.1 (updating the baseline), Topic 5

A reference value is never a bare number (Table 3.4.2)

Field	Meaning	Lab 2 example (SRV-FILE-01)
baseline_id / version	unique name of the baseline and its version; alert rules point at a version	BL-SRVFILE01-LOGONS-v1.0
source	the host, device or account measured	SRV-FILE-01
metric	what is counted and in which unit	successful logons per hour
observation period	start, length, number of observations	2026-09-14 (Monday) + 14 days, 336 hours
kept / excluded	observations used, and the reason for each exclusion	334 kept; 2 excluded (Sept 22 22:00-23:59, contractor migration test)
bucket	the time group the row describes	weekday 09:00
n, mean, std_dev, p95	the statistics of the bucket	10, 15.2, 1.66, 17
warning / alert threshold	candidate boundaries (mean + 2 σ , mean + 3 σ , floor 1, rounded up)	19 / 21
reviewed by / approved by / date	the owner review and the sign-off	R. Tremblay (IT) / owner / 2026-09-28
next review	when the baseline is re-examined	2026-10-16 (Lesson 6.1)

"Sixteen logons" is meaningless. "SRV-FILE-01, successful logons per hour, weekday 09:00, ten weekdays from September 14 to 25, mean 15.2, standard deviation 1.7, baseline v1.0 approved September 28" is a reference value.



3.4.3 Building a baseline: steps 1 to 3

- **Step 1 - choose the metric and the source** from the Lab 1 monitoring plan: you baseline what you decided to watch, in priority order
- **Step 2 - fix the observation (learning) period:** at least two full weeks
 - shorter than a week never sees a weekend; a single week has only one Monday, so nothing can be compared with anything
 - longer is better when you can afford it: a month captures a month-end, a year captures seasons - the pilot cannot wait a year
- **Step 3 - collect, and change nothing on the system during collection:** no new software, users or schedules - otherwise the baseline describes a system that no longer exists
- **The pilot:** SRV-FILE-01, successful logons per hour, Monday September 14 to Sunday September 27, 2026 - 336 hourly counts



Steps 4 and 5: time buckets and statistics

- **Step 4 - time buckets:** hours expected to behave the same way. Tuesday 09:00 compares with Thursday 09:00, not with 03:00 and not with Saturday
 - the pilot: one bucket per weekday hour (24 buckets x 10 values) + one pooled weekend bucket (96 values); a hospital would need different ones - the choice is written down
- Step 5 - three statistics per bucket:
 - mean - the average
 - standard deviation σ (sigma) - how far values usually spread: small σ = predictable, large σ = variable; about 95 % of bell-shaped values fall within 2 σ , 99.7 % within 3 σ
 - 95th percentile p95 - the value below which 95 % of observations fall; not pulled up by one extreme value
- **Weekday 09:00 in the pilot:** $n = 10$, mean 15.2, σ 1.66, p95 17 - "about fifteen logons, rarely more than seventeen"
- **Floor of 1 on σ** (an always-zero hour has $\sigma = 0$); candidate thresholds: warning $> \text{mean} + 2 \sigma$, alert $> \text{mean} + 3 \sigma$, rounded up - the numbers Lesson 3.5 turns into alert rules



Terms to learn - Lesson 3.4

1

Baseline (reference database)

A documented description of the normal activity of a system, measured over a representative period, used as the comparison point for detection.

2

Observation (learning) period

The span of time during which activity is recorded to build the baseline; at least two full weekly cycles.

3

Time bucket

A set of hours expected to behave the same way (for example, weekday 09:00); statistics are computed per bucket.

4

Mean

The average of the values in a bucket.

5

Standard deviation (σ)

How far the values usually spread around the mean; small σ means predictable, large σ means variable.

6

95th percentile (p95)

The value below which 95 % of the observations in a bucket fall; not distorted by one extreme value.

7

Threshold

A boundary derived from the baseline (for example, mean + 3 σ) above which activity is treated as suspicious.



The nine steps of the method

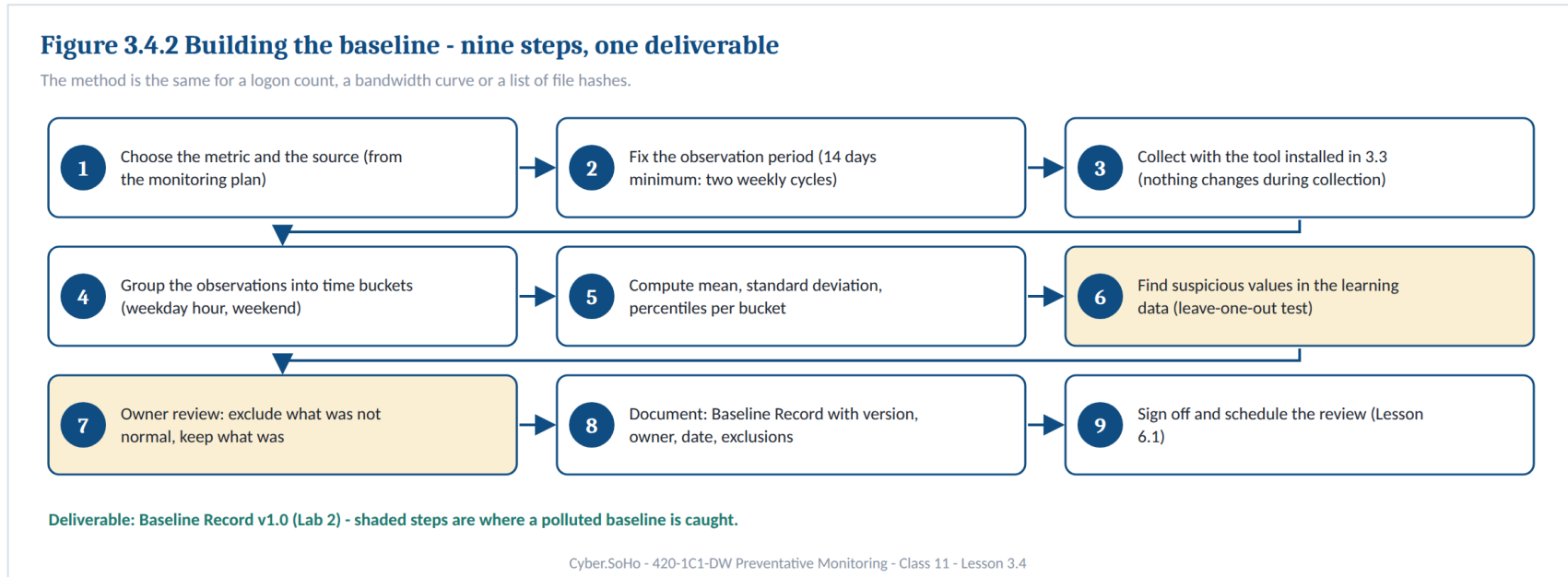


Figure 3.4.2 - Building the baseline in nine steps; the shaded steps are where a polluted baseline is caught.

- Steps 1 to 5 produce numbers; steps 6 and 7 decide whether the numbers describe normal activity; steps 8 and 9 make them a record
- **Was the learning period itself normal?** If an intruder, a contractor's test or a misconfigured backup was active, the baseline learns it as normal and never alerts on it again: the polluted baseline - invisible, because the tool works and the alarm simply never rings

3.4.4 The leave-one-out test and the owner review

- **Step 6 - test the learning data against itself:** each hour is compared with the other hours of its bucket, leaving itself out
 - why leave it out: a value of 35 in a bucket of 0s and 1s would inflate its own mean and σ enough to hide itself (masking)
 - any hour above the mean + 3 σ of its peers is listed as suspicious
- **Step 7 - the owner review is a conversation, not a calculation:** the list goes to the people who know the organization (the owner and the part-time IT contractor)
 - Tuesday, Sept 22, 22:00 (35 logons) and 23:00 (20): a one-time migration test, documented in the change log - excluded
 - Friday, Sept 25, 15:00 (16 logons): a rush before a delivery, confirmed ordinary - kept
- A statistical outlier is a question for the owner, not a conclusion
- **Recomputed on the kept hours:** weekday 22:00 falls from mean 3.7, σ 10.4 (alert boundary 36) to mean 0.22, σ 0.42 (alert boundary 4) - the next migration test at 22:00 will be noticed



Documenting and signing off the Baseline Record

- **Steps 8 and 9 - the record carries:** identifier with version (BL-SRVFILE01-LOGONS-v1.0), period, observations kept and excluded with the reason, bucket definition, statistics per bucket, candidate thresholds, who reviewed, who approved, when, and the next review date
- **The version is not decoration:** when the October 16 review changes a threshold, the record becomes v1.1, and the alert rules of Lesson 3.5 point at a version - "which normal was this alert compared with?"
- **Sign-off by the system owner:** the description of normal is acceptable to the organization - and the owner becomes accountable for the exclusions
- **The FIM (File Integrity Monitoring) baseline follows the same path:** files, SHA-256 values, a date, and the rule that a difference is legitimate only with a change request
- **Two warnings:** a baseline describes the pilot's own systems and nobody else's (Legal Notice 7.2: personal information, kept in the lab, deleted after grading); and it is built from what the tool collected, never from what you believe - the backup at 02:00 stays at 02:00 even if the schedule says 01:00



Table 3.4.3 - Baseline Record v1.0 of SRV-FILE-01 (excerpt)

baseline_id BL-SRVFILE01-LOGONS-v1.0 **period** 2026-09-14 (Monday) + 14 days **excluded** 2 hours (Sept 22 22:00-23:59) **next review** 2026-10-16 (Topic 6.1)

bucket	n	mean	std_dev	p95	warning_threshold	alert_threshold
weekday 00:00	10	0.0	0.0	0	2	3
weekday 02:00	10	1.1	0.3	2	4	5
weekday 07:00	10	5.6	2.01	9	10	12
weekday 08:00	10	13.9	1.92	17	18	20
weekday 09:00	10	15.2	1.66	17	19	21
weekday 12:00	10	8.0	2.14	11	13	15
weekday 15:00	10	12.4	1.62	16	16	18
weekday 17:00	10	5.4	0.66	6	8	9
weekday 22:00	9	0.22	0.42	1	3	4
weekday 23:00	9	0.11	0.31	1	3	4
weekend (any hour)	96	0.07	0.26	1	3	4

11 of the 25 buckets shown; the complete record is Table_3.4.3_Baseline_Record_SRV-FILE-01.csv (companion files) - Section 5 of the Lab 2 documentation.



3.4.5 From baseline to thresholds

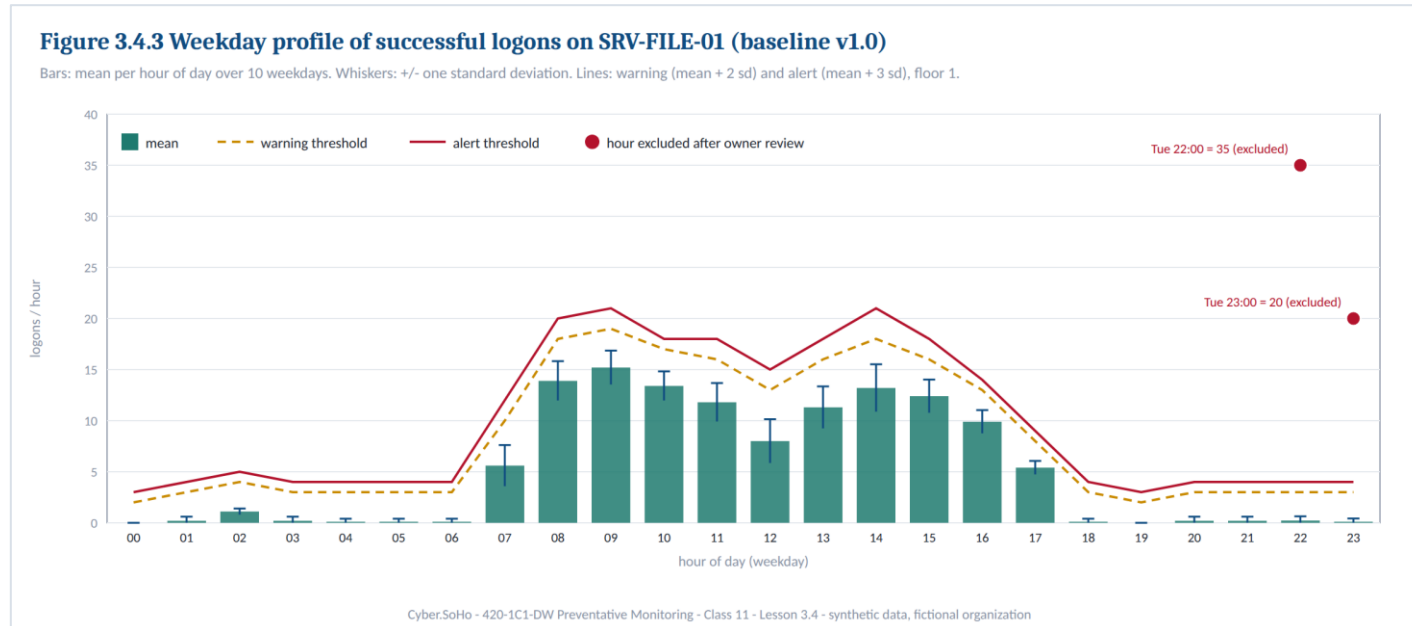


Figure 3.4.3 - Weekday profile of successful logons on SRV-FILE-01: mean, standard deviation, warning and alert boundaries, excluded hours.

- **Weekday 09:00:** mean 15.2, warning above 19, alert above 21. **Weekday 22:00:** mean 0.22, warning above 3, alert above 4
- **Candidate thresholds:** Lesson 3.5 adjusts them for the cost of a false alarm and of a missed event; Topic 6 tunes them again with real alert history - the baseline gives the starting point

Legal Notice checkpoints for Lesson 3.4

Section 6.7 - data and privacy in laboratories

Monitoring collects data, and data about a network is data about people: collect the minimum, in the lab only.

Section 7.2 - monitoring, capture and surveillance data

Baselines, logs and captures come only from Designated Targets in your Authorized Lab Environment and are treated as personal information.

Section 18.2 - submitted work

By submitting, you confirm the work was produced only in the lab and contains no real personal information or real credentials.

Annex A - acknowledgement

The signed acknowledgement must be on file before Lab 2 starts.

Full text: [cyber-soho.github.io/prevmon - Legal Notice, Terms of Use and Acceptable Use Policy \(v2.0\)](https://cyber-soho.github.io/prevmon - Legal Notice, Terms of Use and Acceptable Use Policy (v2.0)) [VirusTotal check](#)



Videos for Lesson 3.4



Secure Baselines - CompTIA Security+ SY0-701 - 4.1 (Professor Messer, 4:11)

Video: <https://www.youtube.com/watch?v=BWPJD9Eb9iE> VirusTotal check · Course page: professormesser.com/security-plus/sy0-701/sy0-701-video/secure-baselines-sy0-701/ VirusTotal check

Establishing, deploying and maintaining a baseline, and checking that systems still match it: the configuration family of Figure 3.4.1 and the review of Lesson 6.1.



Configuration Management - CompTIA Network+ N10-009 - 3.1 (Professor Messer, 5:19)

Video: <https://www.youtube.com/watch?v=cbHcjzLLjuc> VirusTotal check · Course page: professormesser.com/network-plus/n10-009/n10-009-video/configuration-management-n10-009/ VirusTotal check

Production, backup and baseline configurations, integrity checks against a golden configuration, and updating the baseline after a change: the 'configuration and files' family and the versioning of the Baseline Record.



Logs and Monitoring - CompTIA Network+ N10-009 - 3.2 (Professor Messer, 13:06) - optional re-watch

Video: <https://www.professormesser.com/network-plus/n10-009/n10-009-video/logs-and-monitoring-n10-009/> VirusTotal check · Course page: professormesser.com/network-plus/n10-009/n10-009-video/logs-and-monitoring-n10-009/ VirusTotal check

The minutes on flow data and the network performance baseline illustrate the 'network' family: knowing the normal utilisation of a workday before troubleshooting.

Open the VirusTotal check first, then the link (Legal Notice, section 14.2 and Annex C). Third-party content used for identification and educational purposes only (section 5.2).



Real-life example 3.4 (1/2): the first baseline of SRV-FILE-01

- **Task:** Build the Baseline Record v1.0 of SRV-FILE-01 for the metric 'successful logons per hour' from a 14-day export, clean it after the owner review, derive the candidate thresholds, and add a file-integrity baseline of three critical files.
- **Relation to the lecture:** the nine-step method of Figure 3.4.2 on the Tier 1 file server, with the statistics of 3.4.3 and the cleaning of 3.4.4 - on the data collected by the tool installed in example 3.3
- **Deliverable:** the Baseline Record (Table 3.4.3, CSV) and the file-integrity baseline (Table 3.4.4, CSV) = Section 5 of the Lab 2 documentation; the Lab 2 guide supplies a synthetic 14-day export; the review of October 16 updates it to v1.1
- **Why this metric:** the Lab 1 plan lists SRV-FILE-01 as Tier 1 (accounting files and workshop drawings) and "successful logons per hour" as a watched metric
- **The export:** Monday, September 14 to Sunday, September 27, 2026 - 336 hourly counts; business hours 07:00-17:00 carry 5 to 16 logons, nights and weekends almost none, except the backup service at 02:00 every day (expected)



Real-life example 3.4 (2/2): from the export to the signed record

- **First statistics show something wrong:** weekday 22:00 has mean 3.7 and σ 10.4 - no ordinary evening produces that
- **Leave-one-out test lists three hours:** Tue Sept 22 at 22:00 (35) and 23:00 (20); Fri Sept 25 at 15:00 (16)
- **Owner review, Monday September 28:** the contractor recognizes his migration test (a script opening a session per folder, written in the change log) - both hours excluded; the Friday rush is kept
- **Recomputed on 334 hours:** weekday 22:00 returns to mean 0.22, alert boundary 4
- **The record:** BL-SRVFILE01-LOGONS-v1.0, period, 334 kept / 2 excluded with the reason, 25 buckets with mean, σ , p95, warning, alert; reviewed and approved September 28; next review October 16
- **FIM baseline added:** the hosts file, the nightly backup script, the share-permissions file - each with its SHA-256; a change is legitimate only through a change request
- **Tomorrow:** the warning and alert columns become the numbers inside the first alert rules



- SRV-FILE-01 - successful logons per hour - building baseline v1.0** stage: first statistics

14-day export (hourly counts); weekend shaded; the backup logon at 02:00 every day is normal

Weekday profile, raw data (mean per hour, whisker = 1 sd)

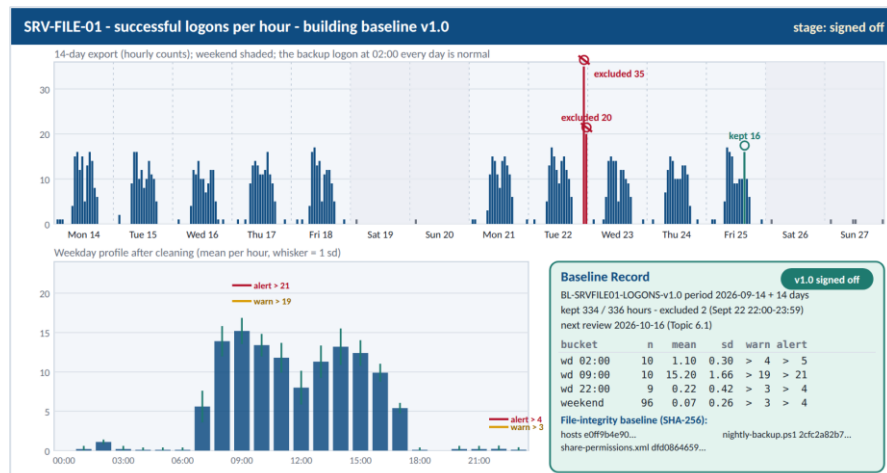
bucket	n	mean	sd	warn	alert
00:00	1	0.5	0.5		
01:00	1	0.5	0.5		
02:00	1	0.5	0.5		
03:00	1	0.5	0.5		
04:00	1	0.5	0.5		
05:00	1	0.5	0.5		
06:00	1	0.5	0.5		
07:00	1	0.5	0.5		
08:00	1	0.5	0.5		
09:00	15	15.0	2.0		
10:00	15	15.0	2.0		
11:00	14	13.0	2.0		
12:00	12	12.0	2.0		
13:00	11	11.0	2.0		
14:00	13	13.0	2.0		
15:00	12	12.0	2.0		
16:00	10	10.0	2.0		
17:00	6	6.0	2.0		
18:00	1	0.5	0.5		
19:00	1	0.5	0.5		
20:00	1	0.5	0.5		
21:00	4	4.0	4.0		
22:00	2	2.0	2.0		

Code example 3.4 (2/2): review, thresholds, file integrity

```
[3] 3 suspicious hours found in the learning data:
    review Tue day 09 22:00 count= 35 -> EXCLUDE - contractor migration test, one-time, documented
    review Tue day 09 23:00 count= 20 -> EXCLUDE - contractor migration test, one-time, documented
    review Fri day 12 15:00 count= 16 -> KEEP - confirmed normal by the owner

[4] statistics recomputed on 334 kept observations
    weekday 02:00      n=10  mean= 1.10 sd=0.30 p95=2   warn>4   alert>5
    weekday 09:00      n=10  mean=15.20 sd=1.66 p95=17  warn>19  alert>21
    weekday 22:00      n=9   mean= 0.22 sd=0.42 p95=1   warn>3   alert>4
    weekend (any hour)  n=96  mean= 0.07 sd=0.26 p95=1   warn>3   alert>4

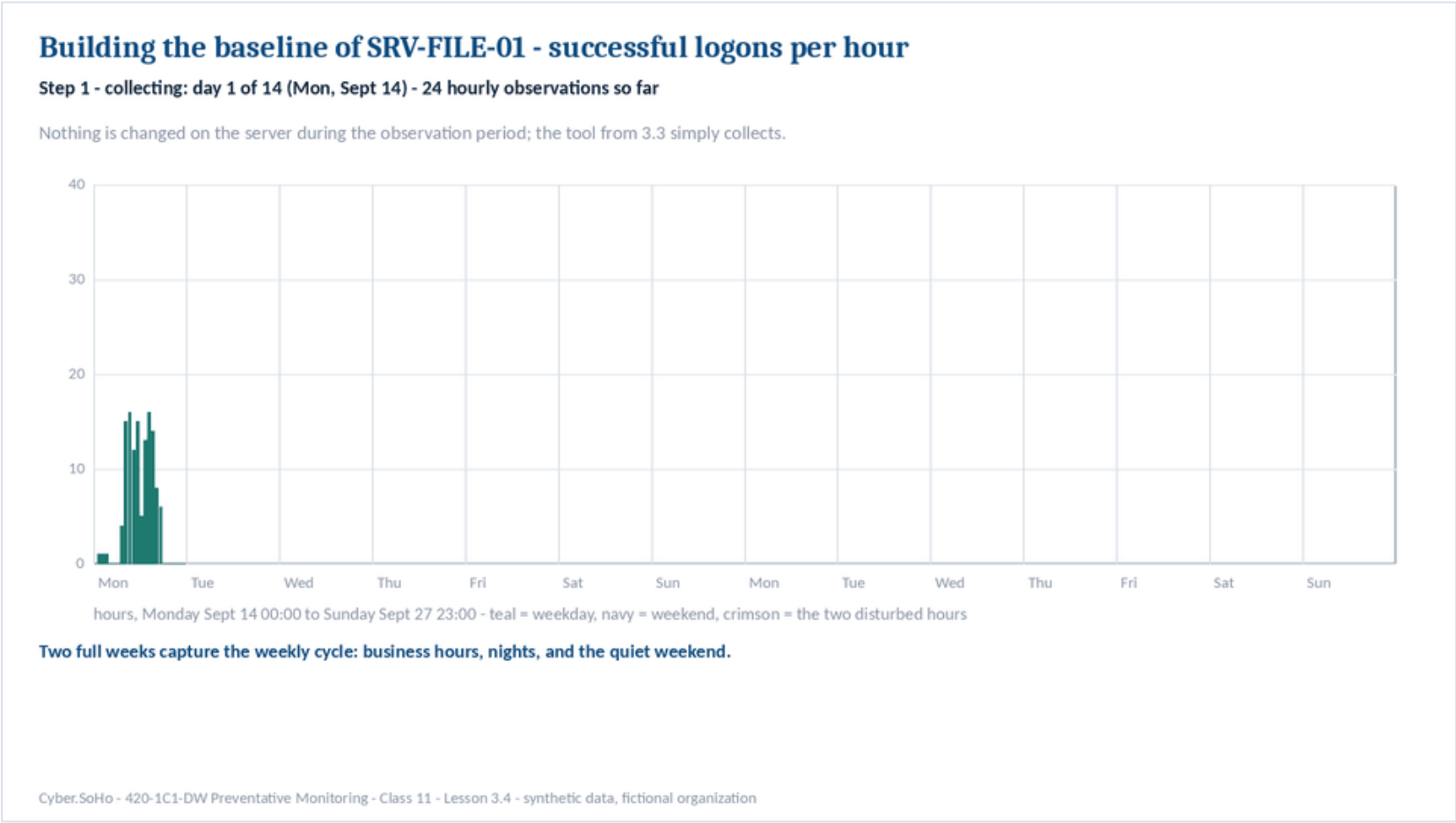
[5] Baseline Record written -> Table_3.4.3_Baseline_Record_SRV-FILE-01.csv
    FIM C:\Windows\System32\drivers\etc\hosts      e0ff9b4e90dc2192...
    FIM C:\Scripts\nightly-backup.ps1             2cfc2a82b70cf5fd...
    FIM C:\ProgramData\VdP\share-permissions.xml  dfd086465980c927...
```



- Three suspicious hours; two excluded with the reason, one kept
- Statistics recomputed on 334 kept observations; four key buckets printed with warning and alert boundaries
- Baseline Record written, then the SHA-256 of three critical files (the FIM baseline)
- Last line: v1.0 ready for sign-off, thresholds feed Lesson 3.5, review on October 16



Animation 3.4 - Building the baseline



Animation 3.4 - The baseline of SRV-FILE-01 built day by day: collection, buckets and statistics, suspicious hours, owner review, thresholds, and the signed-off record.



Table 3.4.4 - File-integrity baseline of SRV-FILE-01

file	sha256	baseline_date	expected_change
C:\Windows\System32\drivers\etc\hosts	e0ff9b4e90dc21925a31793aaf63c22d7440fa04dfabd0854ef76d7e12472bc5	2026-09-27	only through a change request
C:\Scripts\nightly-backup.ps1	2cfc2a82b70cf5fd4febed380cba03bdd6cbba489abc96efea2ace44a388ffce	2026-09-27	only through a change request
C:\ProgramData\VdP\share-permissions.xml	dfd086465980c9273bbc922476358d31b8804d8f76ff0859a9632e2b7ef13e8d	2026-09-27	only through a change request

- The agent computes the hash during its first scan and re-checks it afterwards; the reference list is yours - only you know which files matter
- A difference is legitimate only when it corresponds to a change request; otherwise it is an alert (Lesson 3.5)
- Same function as the download check of Lesson 3.3: SHA-256 (Secure Hash Algorithm, 256-bit)



Flash cards - Lesson 3.4

What is a baseline (reference database)?

A documented description of the normal activity of a system, measured over a representative period, stored with source, metric, bucket, period, statistics, date, owner and version, and used as the comparison point for detecting anomalies.

Why must the observation period cover at least two full weeks?

Because most organizations follow a weekly cycle: a shorter period never sees a weekend, and a single week has only one Monday, so nothing can be compared with anything.

What is a time bucket, and why does the pilot use one bucket per weekday hour plus one weekend bucket?

A set of hours expected to behave the same way. Nine in the morning on a Tuesday compares with nine on a Thursday, not with three in the morning or Saturday; the idle weekend is pooled into one bucket for reliability.

What is a polluted baseline, and how is it prevented?

A baseline that learned abnormal activity from the learning period as if it were normal, so it never alerts on it again. Prevented by testing each hour against its peers (leave-one-out) and by an owner review that excludes what was not normal.

How are the candidate thresholds derived from a bucket's statistics?

Warning = mean + 2 standard deviations, alert = mean + 3 standard deviations, with a floor of 1 on the standard deviation, rounded up; Lesson 3.5 adjusts them and Topic 6 tunes them.



Self-check 3.4 (1/3)

MCQ - one answer

Q7. SRV-FILE-01 shows 35 successful logons at 22:00 on a Tuesday. What decides whether this is an anomaly?

- A)** The number 35 itself: more than 30 logons in an hour is always abnormal.
- B)** The comparison with the baseline value of the weekday 22:00 bucket and its threshold.
- C)** The type of server: file servers are never accessed at night.
- D)** The vendor's default rule set, which knows the normal values of every organization.

Hint: Think about what the tool needs before it can call anything abnormal.

MCQ - one answer

Q8. The weekday 03:00 bucket of SRV-FILE-01 has a mean of 0 and a standard deviation of 0. What does the pilot do before computing the alert boundary?

- A)** Removes the bucket, because a bucket with no activity cannot be monitored.
- B)** Applies a floor of 1 to the standard deviation, so the alert boundary becomes 3 logons.
- C)** Merges the 03:00 bucket with the 09:00 bucket to obtain a larger standard deviation.
- D)** Uses the 95th percentile of the 09:00 bucket instead.

Hint: Mean plus three times zero is zero.



Self-check 3.4 (2/3)

MCQ - one answer

Q9. Why is each hour tested against the other hours of its bucket with itself left out (leave-one-out)?

- A) Because the tool cannot compute statistics on ten values.
- B) Because a very large value included in its own bucket inflates the mean and standard deviation enough to hide itself (masking).
- C) Because the hour under test belongs to a different day of the week.
- D) Because leaving one value out makes the mean equal to the 95th percentile.

Hint: Recall what a value of 35 does to the mean and standard deviation of nine values near zero.

MCQ - one answer

Q10. Which of the following is a complete reference value, as the Baseline Record requires?

- A) "16 logons."
- B) "About 15 logons in the morning on the file server."
- C) "SRV-FILE-01, successful logons per hour, weekday 09:00, Sept 14-25 (10 weekdays), mean 15.2, σ 1.66, p95 17, baseline v1.0 approved Sept 28."
- D) "SRV-FILE-01 is busy from 08:00 to 17:00."

Hint: A reference value carries source, metric, bucket, period, statistics and administrative fields.



Self-check 3.4 (3/3)

MSQ - select 3

Q11. Which three statements about the owner review are correct? (select 3)

- A)** A statistical outlier is a question for the owner, not yet a conclusion.
- B)** Every flagged hour must be excluded, otherwise the baseline is polluted.
- C)** Each decision is recorded with its reason (for example, 'contractor migration test, documented in the change log').
- D)** The Friday 15:00 rush before a delivery was kept because the owner confirmed it as ordinary activity.
- E)** The review is done by the tool automatically from the vendor's rules.
- F)** The review is skipped when the learning period is longer than two weeks.

Hint: The review is a conversation with reasons, and its outcome is not always 'exclude'.

MSQ - select 3

Q12. Which three items must a Baseline Record contain for its numbers to be usable? (select 3)

- A)** A version number and the date of approval.
- B)** The observation period and the number of observations kept and excluded, with the reason for each exclusion.
- C)** The personal names of every employee whose logons were counted.
- D)** The next review date.
- E)** A screenshot of the vendor's website.
- F)** The password of the monitoring server's administrator account.

Hint: Without these, nobody can answer 'which normal was this alert compared with?'



Lab 2 summary (1/2): what Saturday's lab asks

- **Lab 2** = Integrative Project, Part 2 (5 % of the final grade), assigned Saturday, October 3 (Class 12) right after Lesson 3.5
- **Task:** install and configure a monitoring tool on the Part 1 network, build the baseline, and set the alerts - on the Val-des-Pins network you mapped in Lab 1, with your required variation
- **The tool is the one your Class 8 decision selected:** the pilot uses Wazuh on MON-01; a decision record that selected the runner-up installs that one and documents the same stages
- Six sections of documentation:
 - 1-3 Tool Selection Record (Class 8): coverage map, decision matrix, decision record
 - 4 Installation and Update Record (today, Lesson 3.3 - Table 3.3.5 is the model)
 - 5 Baseline Record and file-integrity baseline (today, Lesson 3.4 - Tables 3.4.3 and 3.4.4)
 - 6 Alert rules (Saturday, Lesson 3.5)
- Lab 3 (October 16, Topic 6) reopens Sections 4 and 5 to update the baseline and propose optimizations



Lab 2 summary (2/2): what to bring, and the rules

- **Bring on Saturday:** the lab VM with a fresh snapshot, your Class 8 decision record, the Class 7 priority order, the Lab 1 network map
- **Submission format, file names and due date:** stated in the Lab 2 guide and on the Léa DropBox (Omnivox). Lab 1 is due Sunday, October 11, 23:59 EDT
- Rules that apply:
 - Annex A acknowledgement signed and on file before you touch the lab
 - installation only inside your Authorized Lab Environment, isolated, snapshot first (section 6.4)
 - every download from the official publisher, checked (Annex C, section 14.2)
 - logs, captures and baselines are fictional data kept in the lab, deleted after grading (6.7, 7.2, 18.2)
 - course credentials, virtual machines and datasets are never shared or published (17.2)

Full text

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Wrap-up: what to keep from today

- **Installation is a procedure with seven stages;** preparing, verifying and recording make it trustworthy and repeatable
- **Three kinds of updates, a fixed order (manager, agents, content), never without a way back,** always with a verification that triggers a rollback when it fails
- **A baseline is a documented, versioned description of normal,** built over at least two weekly cycles, cleaned after an owner review, turned into warning and alert thresholds
- Dates:
 - Saturday, October 3 - Lesson 3.5 (alerts) and Lab 2 start
 - Saturday, October 10 - Quiz 3
 - Sunday, October 11, 23:59 EDT - Lab 1 due on the Léa DropBox
 - Friday, October 16 - Lab 3: the baseline built today is reviewed and updated



All references (1/2)

1. Wazuh wazuh.com [VirusTotal check](#)

Publisher of the open-source security platform selected in Class 8 for the pilot; the only place its software is downloaded from.

2. Wazuh documentation - Quickstart documentation.wazuh.com/current/quickstart.html [VirusTotal check](#)

System requirements, the installation assistant, the supported operating systems and the advice to disable the package repository after installing.

3. Graylog graylog.org [VirusTotal check](#)

Publisher of Graylog Open, the runner-up of the Class 8 selection.

4. Ubuntu Server ubuntu.com/server [VirusTotal check](#)

The operating system of MON-01 (24.04 LTS, five years of security updates).

5. VirtualBox www.virtualbox.org [VirusTotal check](#)

The virtualization software of the lab; the 'Take snapshot' action is the way back before every installation and update.

6. Semantic Versioning semver.org [VirusTotal check](#)

The MAJOR.MINOR.PATCH version numbering convention used to read release notes.

7. CVE Program www.cve.org [VirusTotal check](#)

The public registry of identifiers for known software weaknesses (CVE, Common Vulnerabilities and Exposures).

8. NIST National Vulnerability Database nvd.nist.gov [VirusTotal check](#)

Scores and details for each CVE entry, used to decide how urgent a patch is.

9. CISA Known Exploited Vulnerabilities catalogue www.cisa.gov/known-exploited-vulnerabilities-catalog [VirusTotal check](#)

The list of weaknesses that attackers are actively using; an entry in it turns a scheduled patch into an urgent one.

10. Canadian Centre for Cyber Security - ITSM.10.096 cyber.gc.ca/en/guidance/top-10-it-security-action-items-no2-patch-operating-systems-and-applications-itsm10096 [VirusTotal check](#)

Canadian guidance on patching operating systems and applications (action item 2 of the top ten).



Open the VirusTotal check first, then the address (Legal Notice, section 14.2 and Annex C)

All references (2/2)

11. NIST SP 800-40 Rev. 4 doi.org/10.6028/NIST.SP.800-40r4 [VirusTotal check](#)

Guide to enterprise patch management planning: why patching is preventive maintenance and how to organize it.

12. NIST SP 800-92 doi.org/10.6028/NIST.SP.800-92 [VirusTotal check](#)

Guide to computer security log management: log analysis depends on knowing what normal looks like.

13. NIST SP 800-94 doi.org/10.6028/NIST.SP.800-94 [VirusTotal check](#)

Guide to intrusion detection and prevention systems: anomaly-based detection compares activity with a profile of normal behaviour.

14. NIST SP 800-128 doi.org/10.6028/NIST.SP.800-128 [VirusTotal check](#)

Guide for security-focused configuration management: the configuration baseline and its controlled changes.

15. CIS Benchmarks www.cisecurity.org/cis-benchmarks [VirusTotal check](#)

Consensus hardening guides per operating system and application; the longer list behind the hardening stage.

16. Professor Messer - Security+ SY0-701 and Network+ N10-009 courses www.professormesser.com [VirusTotal check](#)

The free video courses the lesson videos come from (Mitigation Techniques, Vulnerability Remediation, Secure Baselines, Configuration Management, Logs and Monitoring).

17. CompTIA www.comptia.org [VirusTotal check](#)

The certification body whose Security+ and Network+ objectives the videos follow.

18. VirusTotal www.virustotal.com [VirusTotal check](#)

The free scanning service used for the link and download checks required by Annex C.

19. Dawson College www.dawsoncollege.qc.ca [VirusTotal check](#)

The college offering the LEA.D8 program; the Léa DropBox on Omnivox is where labs are submitted.

20. Cyber.SoHo Legal Notice, Terms of Use and Acceptable Use Policy v2.0 cyber-soho.github.io/prevmon/00.01.%20cyber.soho%20-%20legal.html [VirusTotal check](#)

The document that governs the use of this lecture, the lab and every deliverable (sections 4, 6, 7, 14, 17 and 18, Annexes A to C).



Videos used in this lecture

1. Mitigation Techniques - SY0-701 2.5 (6:51)

<https://www.youtube.com/watch?v=Fc8ZJfmapbl> VirusTotal check

2. Vulnerability Remediation - SY0-701 4.3 (8:45)

<https://www.youtube.com/watch?v=P9xakfmX70c> VirusTotal check

3. Secure Baselines - SY0-701 4.1 (4:11)

<https://www.youtube.com/watch?v=BWPJD9Eb9iE> VirusTotal check

4. Configuration Management - N10-009 3.1 (5:19)

<https://www.youtube.com/watch?v=cbHczlLjuc> VirusTotal check

5. Logs and Monitoring - N10-009 3.2 (13:06), course page

<https://www.professormesser.com/network-plus/n10-009/n10-009-video/logs-and-monitoring-n10-009/> VirusTotal check

Open the VirusTotal check first, then the link (Legal Notice, section 14.2 and Annex C). No affiliation with, sponsorship by, or endorsement from any vendor, platform or institution (section 5).





See you Saturday: Lesson 3.5 - Configuring alerts, then Lab 2

Bring your lab VM with a fresh snapshot, the Class 8 decision record, the Class 7 priority order and the Lab 1 network map.

Homework: the two Lesson 3.3 videos and the self-check questions of both lessons in the lecture page.

Lab 1 is due Sunday, October 11, 23:59 EDT on the Léa DropBox.

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